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Wave propagation

TFE4130 Exercise 6

Department of Electronic systems

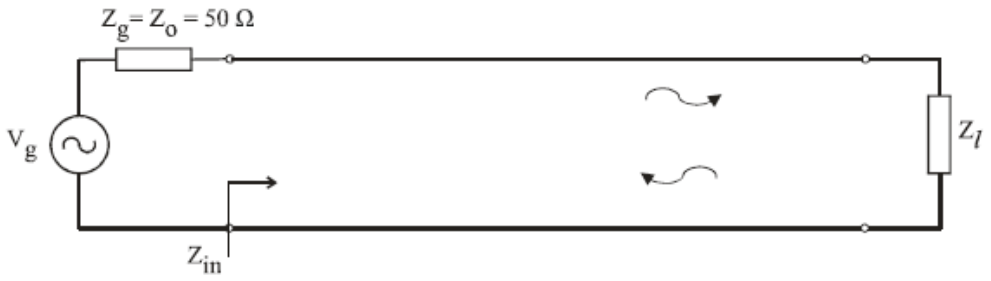
1. Reflections in transmission lines

You want to connect a load Z_L to a transmission line with characteristic impedance Z_0. Assume that $Z_L \neq Z_0$, so if you connect it directly to the line, there will be reflections. These reflections are unwanted, as they may reduce or disturb the signal. How can you eliminate these reflections? Give details of the required component between the transmission line and the load.

2. Transmission line properties

A sine wave generator with frequency ω, impedance $Z_g = Z_0 = 50\Omega$ and idle voltage $V_g = 1V$ is connected to a lossless transmission lin with impedance $Z_0 = 50\Omega$. The line is terminated with a load, $Z_L = 25\Omega$. Under stationary conditions the line voltage will be given by

$$V(z) = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z} \tag{1}$$



a) Explain why the incident voltage wave is described by $e^{-j\beta z}$ and reflected stress wave by $e^{j\beta z}$. Make sure you understand this symbol usage and that you understand how the total voltage and the two voltages waves vary along the line. Both waves have phase velocity $v_p = \omega/\beta$.

b) Assume that the incident voltage wave has amplitude value $|V_0^+| = 1.414V$, ie the effective value of the incident voltage wave is 1 V. At the load, "Ohm Law" requires that $V(0)/I(0) = Z_L = 25\Omega$. (We have chosen to set z = 0 at the load).

i) From this, find V_0^-.
ii) What does that mean that V_0^- is negative in this case?
iii) What is the power delivered to the load P_{avg}, and what is the return loss in dB?

c) The length of the line is $\lambda/10$.

i) What is the reflection factor that the generator "sees"?

ii) What does the phase angle represent for this?
iii) What is the voltage across the generator port?
iv) What impedance does the generator see?
v) Calculate the power that the generator delivers based on the voltage and impedance you found and compare with the previous point.

d) Assume that the line is shorted at the load ($Z_L = 0$) at z = 0.

i) What is in this case the reflection coefficients at positions z = 0, and $z = -\lambda/10$?
ii) What impedance does the generator see now?

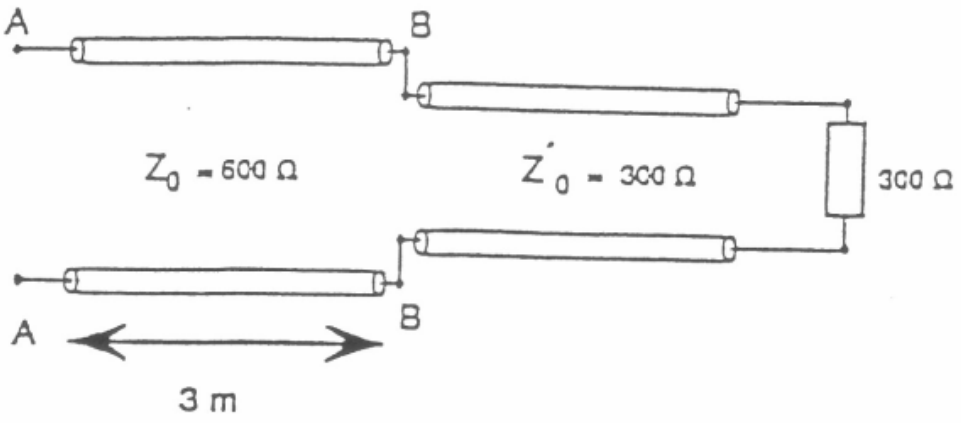
e) Find V_0^- (amplitude and phase) at z = 0, when $|V_0^+| = 2V$ and $\angle V_0^+ = 0°$.

f) What is the total voltage V_L at the load impedance Z_L?

g) Calculate the available power at the generator and the power reflected from the load Z_L.

h) How much power is delivered to the load Z_L?

3. Double transmission line



Consider this double transmission line with characteristic impedances $Z_0 = 600\Omega$ between A-B and $Z_0 = 300\Omega$ from B. $L_0 = 3m$.

a) Find the input impedance as seen from the terminals A - A at the frequency $f = 500MHz$. The phase velocity u_p equals $c = 3 \cdot 10^8 m/s$ for both sections.

b) Repeat for the length A - B equal to $L_0 = 3.15m$.